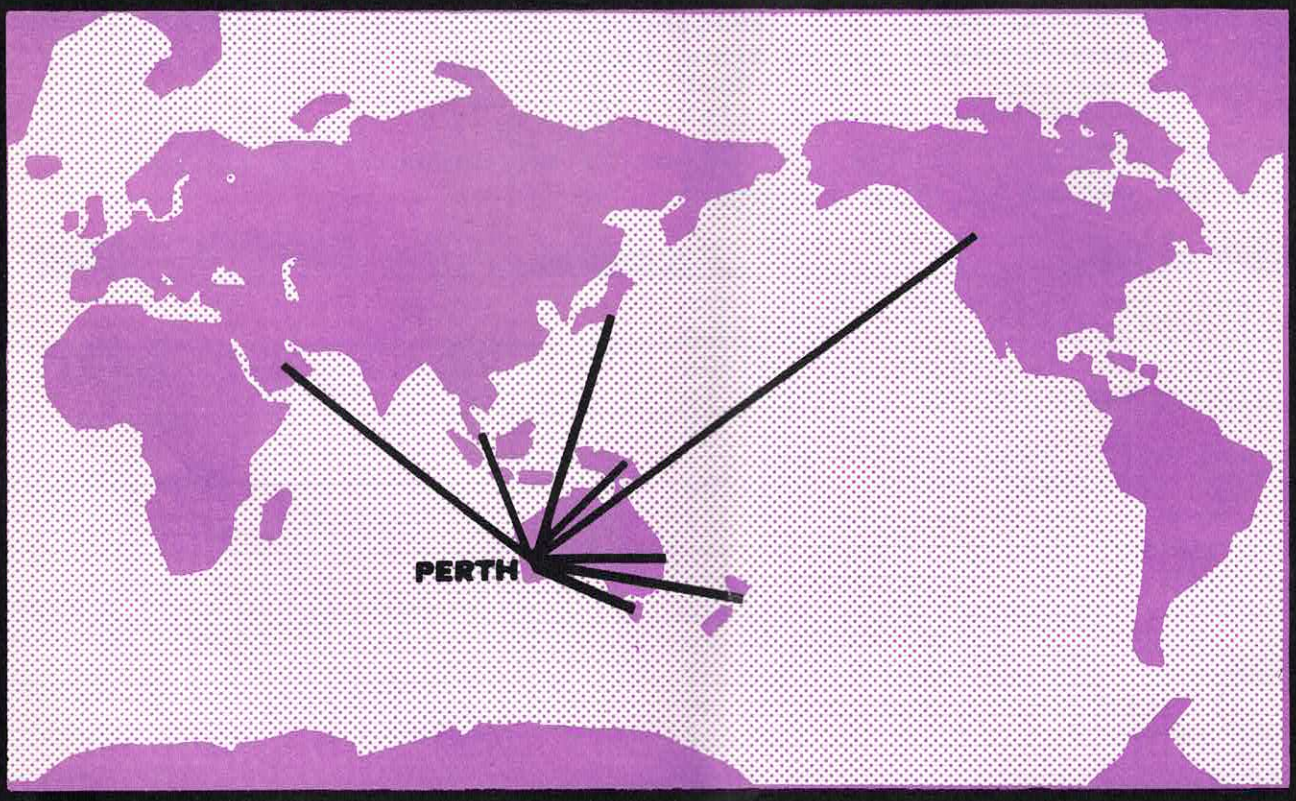
MSX and SPECTRAVIDEO

COMPUTER FORUM

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EDITORIAL

First up I would like to say hello to all our new subscribers. For you who do not know, here in WA ARCHIE MARTINS had a big sale of Sanyo MSX computers and sold 450 of them which meant our little magazine would grow if we could get people to subscribe, it also bought on a bit of a problem, we get 500 magazines run each time in the printers, with this in mind what happens if we have more subscribers than magazines printed. Or do we print 800 magazines and hope everyone subscribes? (More Headaches).

Well I hope everyone had a nice Xmas and New Year, and thankyou to those people who rang us up and wished us the best.

In this issue there is another TUTORIAL but it is only for MSX it has been requested by the new subscribers to the magazine and it starts off very basic and hopefully will finish up teaching people MSX BASIC.

I asked in the last issue for your opinion of the magazine on your likes and dislikes and only recieved 2 replies, so I gather from that most people are happy with what we are doing.

What happened to the front cover ideas?. Maybe people thought we only ran the competition for one issue!. That is not true, please send in your sketch for the front cover to us and if we use it we will send you $10.00 and by the way try to be original some cover ideas we could not use because they were to close to our logo.

One last thing for SVI users I know it seems that you are being left out and it is all MSX but that is not the case. We are trying to get contacts with The Netherlands because the fact that so many SVI machines were sold there, they must have a good following.

I have a favour to ask of any of our readers, if you are going overseas or you have relatives in ENGLAND, JAPAN or THE NETHERLANDS please drop us a line.

ADVENTURES ANONYMOUS



WITH YOUR GUIDE ALAN CRAWFORD

Welcome once again to A.A. This month I will be looking at some new games and supplying some information that may be of interest.

RAINBIRD are releasing a third trilogy of LEVEL 9 adventures. Following JEWELS OF DARKNESS and SILICON DREAMS comes TIME INTO MAGIC. Comprising LORDS OF TIME, RED MOON and PRICE OF MAGIK. Which is great value for the money.

LEVEL 9 have developed a new system for there latest adventures which is called KAOS. This system allows the adventurer a few more options than where seen in their earlier games. New commands include GO TO, RUN TO, FIND (an object), EVERYTHING, EVERYBODY, EXCEPT and FOLLOW (somebody). If for example you typed in GOTO CASTLE you would be taken to the castle and each location that you pass through would be described on the screen as if you had typed in each directional command. It would only stop if something or somebody blocked your path. On the other hand if you typed in RUN TO CASTLE then you would be taken directly to the castle. You can locate objects with the command FIND (object). You can also follow characters that are in the game. If you dont feel like going and getting something, then you can tell one of the characters to do it for you, while you get on with more important business. Although it is not always a good idea as some of the characters are quite stubborn and dont always co-operate in fact DENZYL is the only character in the first part of the game who is controllable. Multi commands are also accepted and in the case of the latest two adventures you can talk to the characters. EG.

TROLL, WAIT 2, IN, KILL VAMPIRE.
MOUSE, WAIT 1, IN, KILL VAMPIRE.
IN, KILL VAMPIRE.

Entering these three commands in sequence outside the vampires lair, brings all three of you into the lair for a simultaneous attack. This seems to be the direction that adventure games are taking. PETE AUSTIN of LEVEL 9 recently remarked, 'The range of puzzles you can have that involve picking up and using objects has been thoroughly explored-introducing characters has to be the next step.

I personely have mixed feeling on the subject, worrying about how to save your own skin is bad enough but to have to depend on a somewhat erratic and mischievous little dwarflike creature for your survival. WELL!!! I ask you? In the case of the mouse, who you are depending upon to kill a vampire, you don't know whether you are getting MICKEY or MIGHTY.

ADVENTURE PREVIEW.

This is a new feature to ADVENTURES ANONYMOUS where I will be looking at brand new games just realeased or about to be released and is a preview as opposed to review. This month I will take a look at the two new releases from LEVEL 9 starting with:-

KNIGHT ORC.

This is a single game which comes in three parts and must be completed in turn to be able to progress to the next stage. With the game comes a 36 page manual containing full instructions, and a novella. It is a good idea to read this novella entitled THE SIGN OF THE ORC, as the games starts where the story ends and it will give you a feel of the characters before you play. The novella incorporates a gay dragon, a hoard of treasure, a tavern and a sense of humour. the three parts of the game comprise-LOOSED ORC, A KIND OF MAGIC and HORDES OF THE MOUNTAIN KINGS. Part 1 of the game is mainly to get you used to the new commands, the more serious stuff comes later. If you like games like THE HOBBIT then you will be at home with this game if not then you had better warm up your joystick.

You play the part of the ORC, GROK GRINDLEGUTS. So to be successful you must act like an ORC. What does an ORC act like you may well ask, according to the manual they are pungent people of elvish stature, with the charisma of blow flies and the appetite of gannets. They are particularly cowardly, too. So you will be doing a lot of running during the game. The ORC philosophy is, 'disgusting is beautiful'. Some of the characters that you will meet are, BOADICEA, ODIN, MUSCLEMAN, THE LIQUIDISER, KRIS and DENZYL.

Cornered in the cellar of the orcs head tavern, a band of orcs has challenged SIR CECIL to a fight of champions. Under the rules as long as the orc champion remains, then the rest may leave, and guess who that is. Thats right GROK GRINDLEGUTS the most drunken orc of the bunch, is left to face SIR CECIL. And just to make sure that you don't loose the spirit of joust, you are tied to your horse.

Meanwhile the band of orcs are being hotly pursued by the ORCS HEAD TAVERN LADIES BOWLING TEAM. During this pursuit the bridge spanning the chasm has been demolished and your job in part 1 is to find enough rope long enough to span the chasm.

Once the joust is over and you are given up for dead it's off you go to find anything that will join together to make an extended rope. Anything will do, like, the halyard on the flagpole, the hawser in the well and various other items that will do the job. As you travel throughout the countryside you will come across hoards of gold treasures, unfortunetly not the object of this game. Except that they may help to distract the locals who are orc hungry. In part 2 the game takes a strange twist, where two parallel worlds exist, the ancient and the modern. But you will have to wait to hear about them, or buy the game.

GNOME RANGER.

I just recieved this game a couple of days before Xmas. So I was pleased to have something new to play over the holidays. GNOME RANGER is the latest release from LEVEL 9 and is an excellent game. It has the same new commands as KNIGHT ORC, of which I found RAM SAVE/RESTORE to be of great use. Also the OOPS command for taking back a move. Handy when you get killed. The game comes with a 48 page book, which is the diary of the main character INGRID BOTTOMLOW. The game commences where the diary finishes, with INGRID being transported to some far of land. She must now make her way back home. The game is in three parts on two cassettes and each part must be completed before you can go on to the next. You will also have to answer a question relating to the diary before you can continue. So you must have the diary to play the game as the questions are picked at random. I'll take a quick look at each part and go into more detail at a later stage. The three parts of the game are (ANIMAL, VEGETABLE and MINERAL). In part one you must defeat the evil witch, with the help of a NYMPH, LEPRECHAUN and a UNICORN. You must help each of the characters before they will help you, and you will need all the help you can get. Points are gained for doing various things and recruiting various characters also for defeating the witch at the end of part one. This part is the easiest of the three.

Part two, and possibly the most difficult of the three parts, is the vegetable part, there are only two vege's to befriend in this part., the TREEHOUSE and the GREENSLAVE. This part consists of making potions out of the various plants that inhabit the garden. These potions must be mixed in the correct order to have the effect they are intended for. You will need help from the vege's to collect some of the ingrediants. I had a few problems with this part of the game, mainly getting the treehouse to follow you. It cannot go where the ground is stoney, so you need to carry the compost heap with you, but it did'nt work all the time, which was a little frustrating.

Part three is the mineral part, four gems must be found and returned to the bronze gate from which they were stolen. You must help the yeti and the penguins and then take on the firepeople and icepeople. The icepeople are the most helpful and once you have returned the kidnapped icechild from the firepeople the icequeen and icejester are invaluable allies. Although the yeti is suppose to help you, he is of no use whatsoever, but you do get points for helping him. The rock garden which is guarded by the gnome, GNUT MYJOB, is a little tricky as you have to give him one of the four gems to enter, but you can't leave if you are carrying anything. But there is a way however and that is via the river. But you will need the help of one of the icepeople.

One complaint is that sometimes the description is to big to fit on the screen in one go, so a press space bar to continue message comes on the screen. That is alright most of the time, but on a few occasions this message has been misplaced and you loose some of the text at the top of the screen. This appears to be a programming error. Otherwise an excellent game.

HAPPY ADVENTURING, ALAN.

THE FOLLOWING FEW PAGES WILL CONCERN ALL THE BEGINNERS IN COMPUTER PROGRAMMING.

First let me make one thing clear BASIC stands for -

[B]iginners [A]ll [P]urpose [S]ymbolic [I]nstruction [C]ode

In this issue we will learn about certain items and their uses.

1.. STRINGS ($)	4.. INPUT	7.. ENTER(return)
2.. VARIABLES	5.. FOR/NEXT	8.. LOCATE
3.. PRINT (?)	6.. REM (')	9.. SAVE/LOAD

Type in this small program,and when you have finished it we will continue.

REM OR ' STANDS FOR REMARK AND WILL NOT BE PRINTED ON THE SCREEN, THEY ARE USED FOR PROGRAMMERS REFERENCE. YOU DO NOT HAVE TO TYPE THEM IN.

AFTER EACH LINE YOU MUST PRESS ENTER OR RETURN

```
10 CLS ' CLEARS THE SCREEN
20 INPUT "WHAT IS YOUR NAME:";NM$
30 INPUT "HOW MANY TIMES SHALL I PRINT IT:";HM
40 FOR X=1 TO HM        'THIS IS THE START OF THE FOR/NEXT LOOP
50 PRINT NM$            'YOU CAN USE ? IN PLACE OF PRINT
60 NEXT X               'THE END OF THE FOR/NEXT LOOP
```

Try to use variables that mean something for example -

AGE$ or AGE (for numbers)
ADRES$ (for street names)
PL1 (for player No1)

Sometimes you may use a Variable such as NAME$ and you will get a SYNTAX error this is because this is a reserved word meaning that the computer uses this word or part of it to perform a particular function as in RENAME, so if this does happen just modify your Variable such as I have for my name (NM$).

Now lets examine it first line 20 INPUT-

Input means that the computer is waiting for you to put in something either a number or letter depending on the VARIABLE meaning either string or numeric.

A STRING or $ VARIABLE can be any key on the keyboard either a number,letter or symbol.
The only key that can not be used is a comma (,) if the STRING is in an INPUT statement, because the comma (,) is used to seperate multiple inputs.

FOR EG-

```
20 INPUT "TYPE IN SEX , AGE:";SEX$,AGE
```

YOU WOULD TYPE MALE,31 [enter]
OR
MALE [ENTER] 31 [ENTER]

LINE 30

SETS UP A COUNTER BY ASKING YOU HOW MANY.

LINE 40

START OF THE LOOP
FIRST TIME THROUGH THE LOOP X=1
NEXT TIME X=2 THEN 3 AND SO ON UNTIL X=HM

LINE 50

PRINTS OUT WHAT IS STORED IN NM$

LINE 60

END OF THE LOOP
IT LOOKS AT X AND IF IT DOES NOT EQUAL HM THEN
IT ADDS 1 TO X AND GOES BACK TO THE START OF THE
LOOP. IN OTHER WORDS X=X+1

Now lets change some things and see what happens.

TYPE IN LINES -

```
15 A$="WELL THAT WAS ":B$="'S I PRINTED "
25 ?:?"WELL ";NM$
35 ?:?"I WILL PRINT YOUR NAME :";HM;" TIMES"
36 ?"AND COUNT IT FOR YOU":?
37 INPUT "PRESS ENTER TO START";ZZ
70 ?:? A$;HM;NM$;B$
```

Now change line 50 to read -

```
50 PRINT X;NM$
```

Now type in run and press enter.

As you can see things have changed a bit just by adding a few lines and changing a few things.
And all the information is stored in what is known as a VARIABLE

```
A$="WELL THAT WAS "      NM$="(WHATEVER YOUR NAME IS)
B$="'S I PRINTED "       HM=(THE AMOUNT OF TIMES TO PRINT)
```

As you can see there are 2 types of variables $(strings) and numeric eg- HM (which only excepts numbers).

You can use numbers in a string($) eg- A$="23 CLIFF STREET", but numeric variables can only contain numbers.

If you wish to save the program you have in your computer onto tape then place a blank tape into your cassette deck, type in CSAVE"NAME" or whatever six letter name you want to save it as, and press [ENTER] then press PLAY & RECORD together, this will save what is in the computers memory onto tape.

When the program is finished saving to tape it is best to VERIFY it by winding the tape back before the start of the program and TYPE in CLOAD? [ENTER] this will check what is on tape and compare it to what is in the computers memory and if they are the same it will say OK else it will give you an error message and if this occurs wind the tape back and resave it.

Once you have finished saving your program type in NEW, this will clear the computers memory. A word of WARNING once you have typed in NEW the program is completely gone so make sure you

save what you want to keep.

Now we can see how we go printing on the screen, type in this small program and run it.

REMEMBER to press [ENTER] or [RETURN] after each line.

```
10 CLS:WIDTH 40:KEY OFF
20 INPUT"TYPE IN YOUR NAME :";NM$
30 LOCATE 0,2:INPUT"PICK A No BETWEEN 1-40 :";A
40 IF A>40 OR A<1 THEN LOCATE 0,2:? SPC(40):GOTO 30
50 LOCATE 0,3:INPUT"PICK A No BETWEEN 1-22 :";D
60 IF D>22 OR D<1 THEN LOCATE 0,3:? SPC(40):GOTO 50
70 CLS
80 LOCATE A,D:? NM$
90 KEY ON:END
```

As you can see it prints your name on the screen using the numbers you type in, now lets see what each line does.

LINE 10

Clears the screen, Sets the WIDTH of the screen and turns the function key display off at the bottom of the screen.

LINE 20

Asks you your name and stores it in NM$

LINE 30

Puts the cursor into location 0 across the screen and 2 down the screen then asks you another input

LINE 40

This line checks to make sure that you do not type in a number that is (>)GREATER than 40 or (<)LESS THAN 1 and if you do it relocates the cursor and prints 40 spaces (SPC) which blanks out that input line and answer.

LINE 50

Same as line 30, but different location and input

LINE 60

Same as line 40, but checks different numbers.

LINE 70

Clears the screen

LINE 80

Set the location of the cursor to (A) across by (D)down and prints your name (NM$).

LINE 90

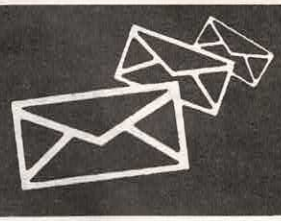
Turns the function keys back on, and then the END

Remember to save this program if you want to keep it, and don't forget to verify what you have saved.

Don't be scared to change the program and explore what different things do, as long as you have the original on tape it does not matter what you do to the program in the computers memory.

Well until next time .. HAPPY COMPUTING.

MAILBOX



Dear Mailbox,

Am enclosing some P.D. stuff for you to use as you see fit. I sure hope this helps some. I have to 'SNEAK' the time for all of this - between work and 2 kids and 'HONEY-DEWS' I have little 'HACKER' time.

Couple of Questions:

1. Where do I get 'WORDSTORE+' and for how much $$?
2. How does one get the function keys on the screen while in CP/M 2.22 80 col. ?
3. Could you provide an address for the SV Distributor in Australia or New Zealand ?
4. Does the expression "YOU AIN'T HAD A GOOD TIME 'TILL YOU PUKE IN YOUR SHOES" really come from Australia??

I really have to disagree with some of Mr. Brisbane's comments (No affence) on SV and MSX. Mainly- SV318 & 328 machines are ''THE'' first MSX machines.

It was after SV did their Copout that the 'Mother' Company 'Bondwell' set up the 'ASCII,Corp' in Japan for the MSX Standard'.

In otherwords, the greed factor by Bondwell (the two main operators originally got filthy rich selling digital watches in the U.S.) was what killed SV. By the way my first 328 was bought in may, 1985 and my sv605 in sept. 1985 and when it broke in feb. 1987 Bondwell told me that it was 'Obsolete' and no parts or service available - I still see red when I think of this BS they sling.

I may yet sue these CHILDREN OF UNMARRIED PARENTS on a Federal level as there are over 5500 SV-328 owners in the U.S.A. and that was another story (how about after the holidays I jot down my 'HISTORY OF FOOBAR' for you? I will format it any way you want, let me know if you do.)

Thanks for your support,
JOHN RUDZINSKI
DETROIT, MICHIGAN, U.S.A.

Thank you for your P.D. stuff. We appreciate the trouble you have gone to and the lack of time available. We, too, have families, jobs and a magazine to put together, so we don't have much time to be 'Hackers' either.

As to where to get 'WORDSTORE+'. As far as we know, it is only available for MSX and comes from AACKOSOFT, Software Distribution, PO BOX 3111, THE NETHERLANDS. You would have to contact them your self to find out the price $$. Does anyone out there know if it is available for the SV-328?

How does one get the function keys on the screen? That's a good question. We will put that to our 'Hacker' and see what comes up. In the mean time, if anyone out there knows the answer, then please drop us a line and we can then pass the information on.

The only address in Australia for the SV that we know of is Greensborough Computers, 67 Grimshaw st. Greensborough Victoria, Australia postcode 3088. In New Zealand we don't know. If anyone can supply us with address's of SV, or MSX, dealers, then please do so. There has been a lot of requests for this information.

As far as the expression goes, I have heard it used but I can't see how that would be a good time, and I don't really know if it originated in Australia although this is the land of rather unusually modified English.

Thanks, John, for your comments on Ross Brisbanes article, and I'm sure Ross put in his articles on the best of his beliefs.

As far as jotting down your History of "Foopar" for us I'm sure that our readers would like to hear from someone from across the waves.

Bye till later.. MAX

Its good to see that someone who is busy at work and at home with a family can sit down for 10 or 20 minutes or even an hour or two and type up some input to the magazine, it allways seems to be the same people that make a real effort to help others and I would just like to say thankyou for your continued support and that others may one day send us some input.

The Editor

Hello Mailbox,

Is there anyone out there that could help? I bought the MSX Computing magazines, typed in and played the games they supplied in the back of the magazine. But the last issue I managed to get, three games I spent hours on I can't get to work. Jump Man, Titan land and MSX muncher. If anyone has these games going and found printing mistakes how about writing in please.

I would like to see more Basic Tutorial in Computer Forum as us older computer buffs didn't learn basic at school. I found the column a great help.

Signed... Still Searching

Well Mr Still Searching, I can not really help you with the programs in MSX Computing that is entirely up to one of our readers but you will notice in this issue we have a brand new Tutorial based on MSX because over here in W.A. just before Xmas there was a big sale of MSX Sanyo machines around 450 to be precise and most of the people have subscribed to our magazine and they asked if we could run a Tutorial for them.

Dear Mailbox,

I have 5 questions to ask,

1 Where was MAILBOX in the last issue.
2 What happened to Robert Brinkworths HACKERS COLUMN.
3 Is there any other English MSX magazine around.
4 Can I buy back issues, I need V1N10 + 11
5 Will you magazine continue or fold up like everything else, the reason I ask is because your last Editorial sounded very depressing.

Regards Kevin James Victoria

Well Kevin to answer your questions in sequence, MAILBOX was left out last issue because of 2 reasons.

1. *There was not enough input for 2 full pages of print.*
2. *Because of our magazines size we have to leave out something to get other things in, you will notice in this issue there is no SOFT-ADDS.*

Robert Brinkworth has left us without any warning so his former position is left open if anyone would like to write to us to do his column we would like to hear from you.

There is to our knowledge no other English MSX magazine in the world, but maybe if you write to the user groups you could get there news letters.
Japan has 3 MSX magazines but unfortunately they are all in japanese.

Back issues are available for $3.50 each which includes postage.

I do apologize for my Editorial sounding so depressing, but the main thing is we will be around for a long time yet, the reason I write in the fashion I do is because I believe that we can keep going with peoples help and that is all I ask for.

USER GROUPS

SYDNEY

We have been advised of a group in Sydney for Spectravideo owners- Please contact Malcolm Samuels on 888 7664 after hours.

QUEENSLAND

BRISBANE SPECTRAVIDEO & MSX USERS GROUP
C/- Mrs Lucille Parker(Secretary)
25 Primrose St.
Woodridge,Queensland 4114
---- Phone (07)208 5951 ----

VICTORIA

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48 Shawlands Avenue
Blackburn South Victoria 3130
---- Phone (03)878 6169 ----

We meet on the first Saturday afternoon of each month (except January) at the Nunawading Library meeting room Whitehorse Rd. Nunawading.

GIPPSLAND MSX USER GROUP
C/- Ross Johnson
---- Phone (051)344 753 ----

WESTERN AUSTRALIA

THE MSX & SPECTRAVIDEO USERS GROUP OF W.A
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FINLAND

SVI/MSX Club/Turku
C/- Mika Ristolainen
Lumikonkatu 7H 272 SF-20240
Turka Finland

If any of the above groups wish to enter their meeting times & place or phone number, drop us a line and we will include them in this page.

SORT Pt 2

by Bruce Goodman

In part one the Bubble Sort technique was discussed, and an SV implementation of the method provided. This issue the 'Insertion' method will be described. The two program segments included here should be added to the code supplied with Part 1.

[The code referred to above was left out last issue,]
[but you will find it in this issue.]
[Max - Assistant ED.]

Unlike the Bubble method which is used to sort an already established list, the principle of the Insertion technique is to continually accept new entries and place them into the list in the correct sequence - thus a sorted list is progressively built up.

List 1 is an implementation of this method. Line 3030 gets the next entry, and Lines 3050 - 3070 searches linearly through the array of numbers already held until the correct location for this number is found. Lines 3080 - 3130 then move the appropriate entries one position to make room for the new one, which is then entered in Line 3140.

A comparison of the times taken to sort say, 100 elements using the Insertion and Bi Directional Bubble methods shows the Insertion technique to be markedly faster. However, there is a cost. Whereas the Bubble methods sort an existing list, the Insertion implementation here requires twice the array space, one array is used to hold the numbers to be sorted and the second to establish the sorted list. (Lines 7070 - 7130 refer) Also as the number of elements to be sorted increases, so does the number of comparisons needed to find the new number's correct position. Further, the quantity of shifts required to create a vacant array space for the new entry is proportional to the sort quantity.

The Binary Insertion method at List 2 aims to reduce the time taken to find the correct position for the current number by the adoption of a 'Binary Search' technique. Essentially this involves comparing the new entry with the one half way down the already established list, and depending on the result, making the next comparison either half way from the middle to the top or bottom of the list. By jumping over a number of array entries in the direction required the total time taken to find the correct position is reduced. (Lines 4050 - 4090 provide this function) However, because this method inherently involves many computer operations, its benefit does not become apparent until the number of items to be sorted exceeds 200. (in this implementation)

Comparison of the four methods described so far is invited. To save a deal of time however, the following typical sort times for 500 elements are provided.

One Way Bubble	43.42
Bi Derectional Bubble	26.40
Insertion	15.00
Binary Insertion	11.39

In the final part of this series two 'LINKED STRUCTURES' techniques will be described.

SORT

```
10 ' SORT  by Bruce Goodman, 26 Sep
 87
20 '
30 GOSUB10000:GOSUB10041:DEFINTD-Z
40 DEFSTRA-C:Z=RND(-TIME):WIDTH39
50 COLOR14,1:SCREEN0,1:CLS:LOCATE,,
0
60 ONINTERVAL=50GOSUB7240
70 LOCATE17,0:PRINT"SORT"
80 GOSUB7330
90 PRINT:PRINT:A=STRING$(9,32)
100 FORI=1TON:PRINTA;B[I]:NEXTI
110 PRINT:PRINT
120 PRINTA;"Press Type Selection"
130 TY$=INPUT$(1)
140 FORI=1TON:IFTY$=CHR$(I+48)THEN1
50ELSENEXTI:GOTO130
150 PRINT
160 PRINTA;"Enter Qty to be Sorted
";
170 INPUTNM:IFNM<2ORNM>500THEN160
180 TY=VAL(TY$):CLS
190 A=MID$(B[TY],4)+" Sort,"+STR$(N
M)+" Elements"
200 LOCATE19-INT(LEN(A)/2),0:PRINTA
210 LOCATE19-INT(LEN(A)/2),1
220 PRINTSTRING$(LEN(A),"«");
230 LOCATE13,22:PRINTCHR$(27)+"p";"
 Time-  0: 0 ";CHR$(27)+"q";
240 ONTYGOSUB1000,2000,3000,4000,50
00,6000
250 GOTO250
990 '====== One Way Bubble ======
1000 GOSUB7010
1010 GOSUB7310
1020 FL=0
1030 FORI=1TONM-1
1040  IFZ[I,1]>Z[I+1,1]
          THEN SWAP Z[I,1],Z[I+1,1
]:                FL=1
1050 NEXTI
1060 IFFL=1THEN1020
1090 N=1:GOSUB7290
1100 GOSUB7220:RETURN
7000 '==== Nos Set Up - Bubbles ===
==
7010 DIMZ[NM,1]:N=1
7020 FORI=1TONM
7030  Z[I,1]=RND(1)*1000+1
7040  GOSUB7440
7050 NEXTI
7060 RETURN
7210 '====== Print Results ======
7220 FORI=1TONM:GOSUB7440:NEXTI:RET
URN
7230 '======== Timer ==========
7240 TI=TI+1:SE=TIMOD60:MI=TI\60
7250 LOCATE19,22:PRINTUSING"###";MI
;
7260 LOCATE23,22:PRINTUSING"##";SE;
7270 INTERVALON:RETURN
7280 '======= Time Off ========
7290 INTERVALOFF:BEEP:FORI=894TO906
:Q=VPEEK(I):VPOKEI,Q+96:NEXTI:RETUR
N
7300 '======== Time On ========
7310 BEEP:FORI=894TO906:Q=VPEEK(I):
VPOKEI,Q-96:NEXTI:INTERVALON:RETURN
7320 '======= Titles ==========
7330 N=0
7340 READB:IFB="End"THENRETURN
7350 N=N+1:B[N]=B:GOTO7340
7360 DATA "1..One Way Bubble"
7420 DATA "End"
7430 '======= Print Nos =======
7440 IFN>100THENRETURN
7450 LOCATE3+((N-1)\20)*7,2+(N-1)MO
D20
7460 PRINTUSING"####";Z[N,1]
7470 N=N+1:RETURN
9999 '========= Begin Module ======
===
10000 ZZ$="SORT.BAS":GOSUB10003
10001 GOSUB10022:STOPON:ONSTOPGOSUB
10020
10002 ONERRORGOTO10038:RETURN
10003 KEYON:ONKEYGOSUB10016,10018,1
0031,10005,10020:RETURN
10004 ON KEY GOSUB 10016,10018,1003
1,10005,10020:RETURN
10005 INTERVALOFF:COLOR1,1,1:SCREEN
0,0:CLS
10006 LOCATE10,10,0:PRINT"Save as "
;ZZ$
10007 LOCATE16,12:PRINT"Y or N":Z$=
""
10008 INTERVALON:ONINTERVAL=25GOSUB
10014
10009 ON INTERVAL=25 GOSUB 10014 :
Z$=""
10010 Z$=INKEY$:IFZ$=""THEN10010
10011 IFZ$="Y"ORZ$="y"THENSAVE"1:"+
ZZ$:INTERVALOFF:COLOR 10,1:CLS:RUN
10012 IFZ$="N"ORZ$="n"THENINTERVALO
FF:RUN
10013 BEEP:GOTO10010
10014 IFZ=0THENCOLOR10:Z=1:ELSECOLO
R1:Z=0
```

```
10015 BEEP:RETURN
10016 GOSUB10022:INTERVALOFF:COLOR1
4,1
10017 SCREEN0,1:LOCATE,,1:LIST-9999
10018 INTERVALOFF:COLOR1,1,1
10019 SCREEN0,0:LOCATE,,0:RUN
10020 COLOR1,1,1:SCREEN0,1:LOCATE,,
1
10021 INTERVALOFF:GOSUB10022:COLOR1
1:END
10022 KEY 1,"CLRLIN"+CHR$(21)
10023 KEY 2," AUTO":KEY 3," LIST"
10024 KEY 4,CHR$(21)+"CLS:LIST"+CHR
$(13)
10025 KEY 5,CHR$(21)+" RUN"+CHR$(13
)
10026 KEY 6," RUN"+CHR$(34)+"1:"
10027 KEY 7,"SAVE"+CHR$(34)+"1:"
10028 KEY 8,"LOCATE,,1"+CHR$(13)+CH
R$(30)+CHR$(21)+CHR$(30)+CHR$(21)+C
HR$(30)+CHR$(21)
10029 KEY 9,"COLOR 2,1,1"+CHR$(13)
10030 KEY 10," FILES"+CHR$(13):RETU
RN
10031 INTERVALOFF:COLOR13,1:SCREEN0
,0
10032 LOCATE10,5,0:PRINT"Total Memo
ry = 21191"
10033 LOCATE11,10:PRINT"Memory Left
 =";
10034 PRINTUSING"######";FRE(0)
10035 LOCATE11,15:PRINT"Memory Used
 =";
10036 PRINTUSING"######";21191-FRE(
0)
10037 GOTO 10037
10038 COLOR10,1:SCREEN0,1:LOCATE9,9
,1
10039 PRINT "Error No";ERR;"in Line
";ERL
10040 GOSUB10022:PRINT:LIST.:END
10041 KEY1," List":KEY2," Again":KE
Y3," Fre(0)":KEY4," Save":KEY5,"  E
nd":RETURN
```

INSERTION

```
2990 '====== Insertion ======
3000 GOSUB7080
3010 GOSUB7310
3020 FORI=1TONM
3030  NO=T[I,1]'  Get the No
3040  IFI=1THENZ[I,1]=NO:GOTO3150
3050  ' Find place to Insert
3060  FORJ=1TOI-1
3070   IFNO>Z[J,1]THENNEXTJ:GOTO314
0
3100    'Insert here, move larger N
os
3110    FORK=ITOJ+1STEP-1' Start @
Top
3120     Z[K,1]=Z[K-1,1]'Move each
Up
3130    NEXTK
3140   Z[J,1]=NO' Put new No in
3150 NEXTI:GOSUB7290:N=1
3160 GOSUB7220:RETURN
7070 '==== Nos Set Up - Inserts ===
==
7080 DIMZ[NM,1],T[NM,1]:N=1
7090 FORI=1TONM
7100  Z[I,1]=RND(1)*1000+1
7110  GOSUB7440
7120  T[I,1]=Z[I,1]:Z[I,1]=0
7130 NEXTI:RETURN
7380 DATA "3..Insertion"
```

BINARY INSERTION

```
3990 '====== Binary Insertion =====
=
4000 GOSUB7080
4010 GOSUB7310
4020 FORI=1TONM
4030  NO=T[I,1]'  Get the No
4040  IFI=1THENZ[I,1]=NO:GOTO4150
4050  ' Find place to Insert
4060  PT=I:PB=1
4070  PC=PB+(PT-PB)/2
4080  IFNO<Z[PC,1]
       THENIFPT=PC
        THENJ=PC:GOTO4110
         ELSEPT=PC:GOTO4070
4090  IFNO>Z[PC,1]
       THENIFPB=PC
        THENJ=PC+1:GOTO4110
        ELSEPB=PC:GOTO4070
       ELSEJ=PC
4100    'Insert here, move larger N
os
4110    FORK=ITOJ+1STEP-1' Start @
Top
4120     Z[K,1]=Z[K-1,1]'Move each
Up
4130    NEXTK
4140   Z[J,1]=NO' Put new No in
4150 NEXTI:GOSUB7290:N=1
4160 GOSUB7220:RETURN
7390 DATA "4..Binary Insertion"
```

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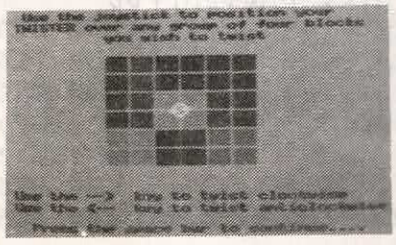
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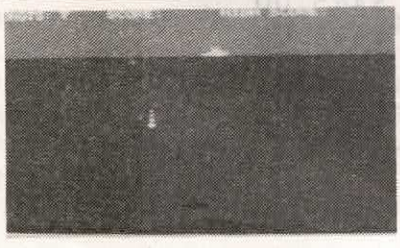
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SVPAK23 - MSXPAK23 -

TWISTER - Like a two dimensional Rubiks cube, this game displays nine different coloured squares each containing four smaller squares. These 36 squares are then jumbled around on the screen and you must rebuild the original display. Its very simple - VERY addictive!

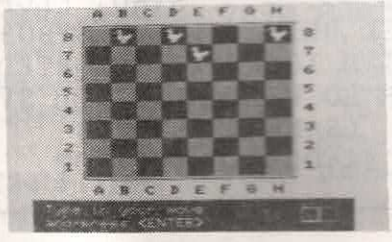
REVERSE - Your task is to arrange the ten jumbled numbers into ascending order by reversing some or all of them. Check it out - its not easy!



TS852 - MSX852 -

GRAND PRIX - You are in the drivers seat in a race to the chequered flag while avoiding slower cars. Hitting another car or two wheels off the track will slow you down and lose points.

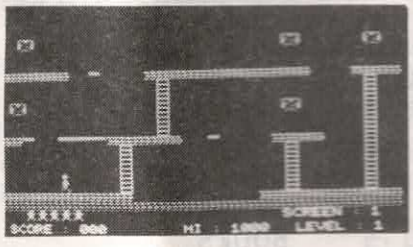
SUBSTRIKE - Drop depth charges on the enemy below. The deeper he is or the faster he moves, the more you will score.



SVPAK24 - MSXPAK24 -

FOX & GEESE - Can you as the Fox penetrate the line of advancing geese and get to the other side of the board before they corner you.

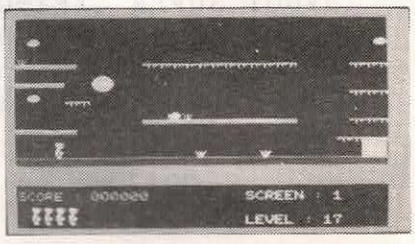
THINK - This is a form of word mastermind. You must guess the word that the computer has selected from the vast number stored in its data banks. This game will certainly make you do as the name suggests.



SVPAK7 - MSXPAK7 -

BOMB SCARE - Very addictive game in which you must climb to different platforms and defuse 5 bombs before your bonus runs out. Has 6 separate screens with the hazards of moving platforms and conveyor belts with holes in them.

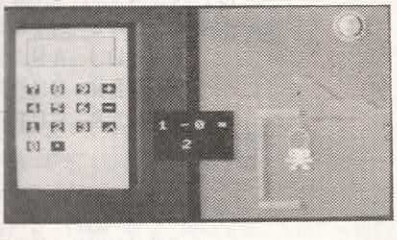
DUNGEON ADVENTURE - A text adventure where you explore the dungeon fighting monsters and gain treasures to discover the location of the great Orb.



SVPAK21 - MSXPAK21 -

GOLDRUSH - Move the miner around the different levels of the mine collecting gold and dodging high explosive charges and falling rocks. The green ledges crumble if you don't get off quickly. When you have collected all of the gold, get to the lift cage for a new screen.

PYRAMID - This is another fascinating text adventure built around the vaults of a pyramid.



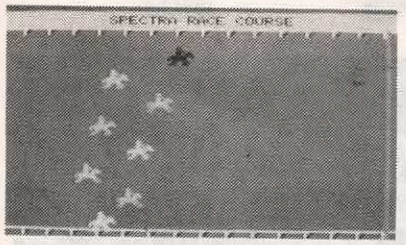
SVPAK25 - MSXPAK25 -

OZMAP - You must drive the car around Australia finding the places required as quickly as you can.

OZQUIZ - Answer general knowledge questions about Australia and her people.

MATH DUEL - Makes practicing maths fun! Has 4 levels of addition and subtraction maths.

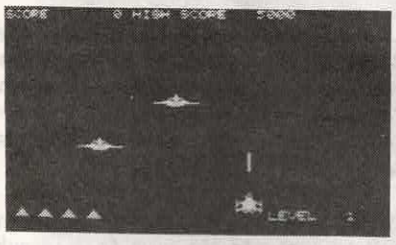
Prices include postage and packing within Australia



SVPAK1 - MSXPAK1 -

SPECTRA DERBY - Place your bets and have a day at the races. Even if you lose its painless.

HANGMAN - The classic word guessing game in which you will be gradually hung for each wrong letter.

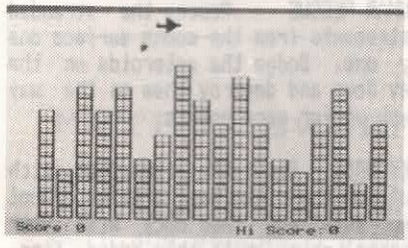


SVPAK10 - MSXPAK10 -

DEMON ATTACK - Shoot the attacking demons as they swoop overhead. You can only destroy a demon when it is fully formed. Killing ten demons gets you on to a harder set.

SEA DIVER - Dodge sea creatures as you make your way to the sea bed. Grab a gold bar and get back to the surface.

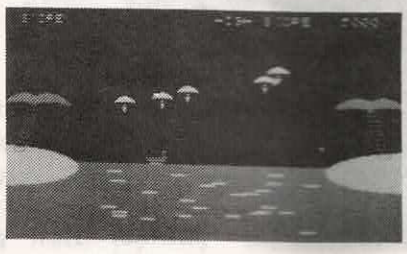
DEATH CITY - Race around the maze trying to shoot your computer opponent but be careful, he is also shooting at you.



SVPAK2 - MSXPAK2 -

SNAKE - Move the snake around to eat the diamonds, but mind you do not cross your own tail.

CITY BOMBER - You must destroy the towering buildings of the city before you can land safely.

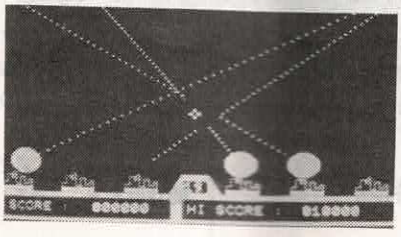


SVPAK8 - MSXPAK8 -

EXPLORER - Watch out for the Pitfalls in this game which will test your control of the joystick. Crocs and other obstacles bar your way as you wander through the jungle in search of treasures.

PARACHUTE - Catch the paratroopers in your boat as they jump from a helicopter into a shark-infested lagoon. The more you catch, the faster they jump.

ANDROID CAVERNS - Move around the caverns of this game and collect the treasure guarded by the androids. But watch out, the androids learn by their mistakes.

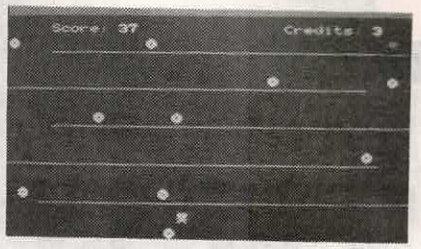


SVPAK22 - MSXPAK22 -

MISSILE COMMAND - Shoot down the incoming warheads before they destroy your cities. You get bonus cities awarded for high scores.

MASTERMIND - Your task is to match the same hidden four-colour combination that the computer has chosen.

LEAKY ROOF - Catch the raindrops leaking through the roof and empty the bucket out of the window before you drown.

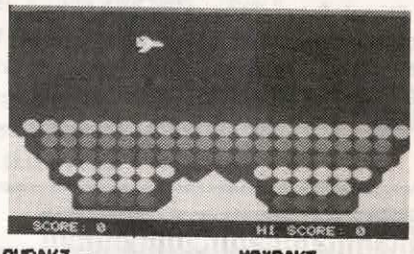


SVPAK4 - MSXPAK4 -

SPECTRA KONG - Win the lady's heart by getting to the top of the screen - but watch out for falling barrels.

TENNIS - A colour version of the original black and white TV game.

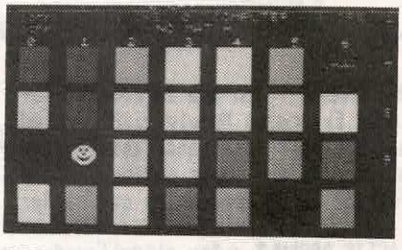
MUSICAL KEYBOARD - This program turns your computer keyboard into a musical keyboard and allows you to select note length and octave.



SVPAK3 - MSXPAK3 -

BOMBS AWAY - Destroy the targets in the ravine below. Be warned - it gets harder the more targets you hit.

BUG CATCHER - Two players move around the garden and the one with the most bugs caught - wins. Great for younger children.

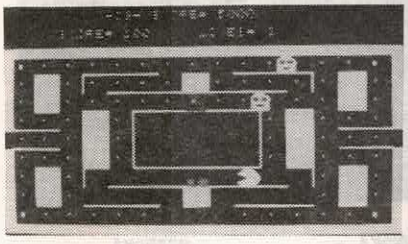


SVEP1 - MSXEP1 -

COPY CAT - The computer will play a set of notes and you must play them back in sequence. There are two levels of play - one note added to the same sequence or a new sequence each time.

MEMORY - A 1 or 2 player game of memory exercise. Uncover two squares on the screen and if the objects revealed match you get another chance

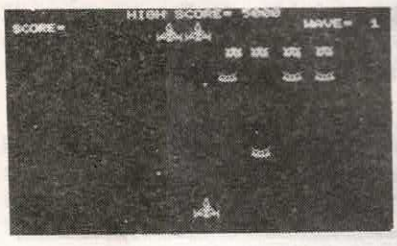
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SVPAK12 - MSXPAK12 -

MUNCH MAN - Guide your Munchman around the maze eating the dots while avoiding the deadly ghosts. If you eat one of the larger dots you will be able to eat the ghosts for a short while. Gain extra points by eating the fruit that occasionally appears on the screen.

ESCAPE FROM PLANET BETA - This is a text adventure where you are stranded on Planet Beta and must find all the necessary equipment to get off before the planet explodes. Only requires one or two-word commands - eg GET LASER.



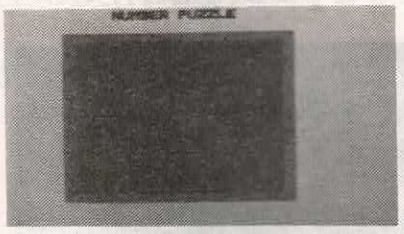
SVPAK5 - MSXPAK5 -

LUNAR LANDER - You are in command of a Lunar Excursion Module with limited fuel but full rotational ability and must land on the lunar surface. Has 8 different landscapes which get harder

GALACTIC ASSAULT - The alien invaders hover above you and one at a time dive on you. Destroy each wave before they get you. For 1 or 2 players.

ROAD RACER - Try and pass as many cars just as fast as you can without crashing.

SPRITE DEFINER - Design sprite shapes on the screen and rotate them when you have finished. Has grids for both sprite sizes.



SVPAK20 - MSXPAK20 -

ARTILLERY FIRE - A two player game where each can select a firing angle and velocity to lob shells on his opponent.

NUMBER PUZZLE - Computerised version of the popular hand-held game in which you must slide numbers around into ascending order.

TOWERS OF HANOI - Move all of the discs from peg 1 to peg 3 without placing a large disc on a smaller one.

TENNIS - This is a two player game of Table Tennis.



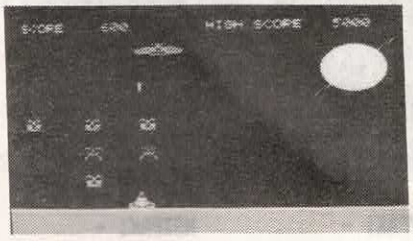
SVPAK6 - MSXPAK6 -

HOPPER - Help Hoppy get across the road (SPLAT!) and then to cross the river (SPLOSH!) and if you are lucky he won't get eaten by the crocs (GULP!!).

SPACE WAR - Challenging game for two players in this duel between your spaceship and that of your opponent. First one to ten points wins.

OTHELLO CHALLENGER - The classic mind game where you must outflank your opponent to capture his pieces. For 1 or 2 players.

PAINTER - Explore the flexibility of your computers colours with this on-screen paint brush.

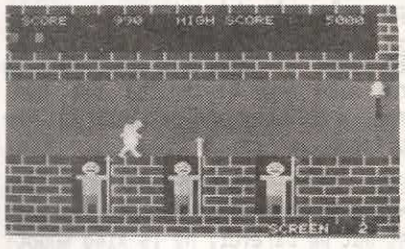


SVPAK9 - MSXPAK9 -

SPACE INVADERS - Shoot the aliens before they invade your planet. Move your ship left and right and shoot missiles at the aliens. Once they have landed the game will end. Score extra points by shooting the UFO which flies overhead.

SUB HUNT - Drop depth charges on the attacking subs and dodge the missiles they fire. You can fire two depth charges at once by firing while moving left or right. Each of the subs is marked with its value.

CHICKEN - Who cares why the chicken crossed the road, we have to find out how! Each time you make it across, the cars speed up.



SVPAK11 - MSXPAK11 -

LUNAR RESCUE - Rescue the stranded astronauts from the moons surface one by one. Dodge the asteroids on the way down and destroy them on the way back up for more points.

SKY BASE - Blast the aliens but watch out for their kamikazi attacks. Shoot two sets and then dock with your mother ship. With this added firepower go on to destroy the alien base.

QUASIMODO - Can you as the hunchback Quasimodo rescue the beautiful maiden from the French soldiers. Cross the screen avoiding fireballs and spears, then ring the bell to get the next screen.

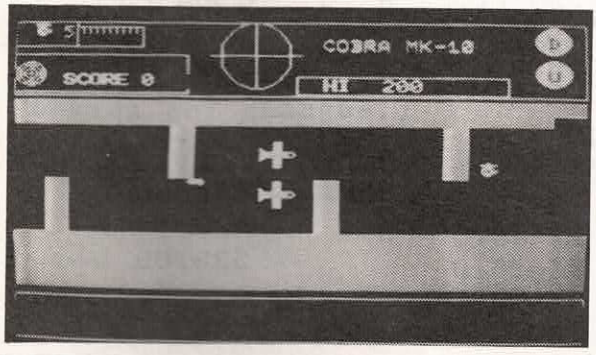
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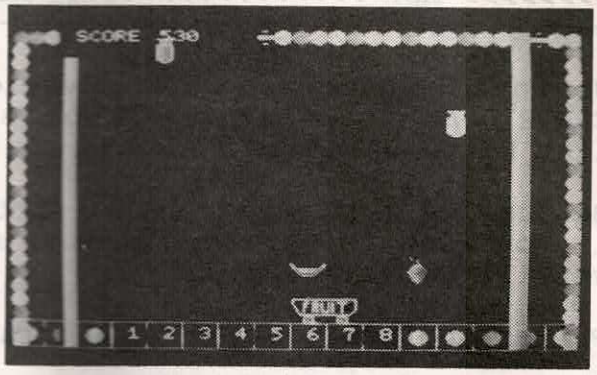
SVPAK26 - MSXPAK26 -

JUMPING COWBOY - A 'platform' game of nine screens where you must collect all the opals to get the next screen.

SUPER SMASHER - Move the bat to smash the ball through the two brick walls. Has three levels of difficulty.

FRUIT CATCHER - Catch falling fruit in your cart. As you catch more, the types of fruit will change and fall at a faster rate.





SVPAK27 - MSXPAK27 -

MIRKWOOD FOREST - Graphics adventure where you must escape from the forest by finding four keys, eating food to maintain energy, while avoiding the spiders. This one will really test your adventure gaming ability.

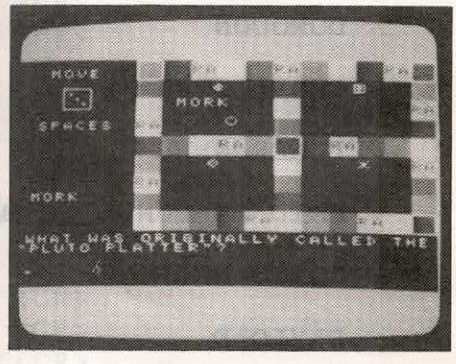
CHOPPER PILOT - The fourteen levels of this game make it very compelling. Keeping clear of canyon walls, fly your helicopter to do battle with enemy fighters, bombers, missiles and UFOs. Great skill required.

UFO (MSX ONLY)

In this machine code game, you are under attack from UFOs. Watch out for the alien unit moving across the bottom of the screen - this will fire 'smart bombs'. Watch out for the meteor shower. Refuel when the truck appears on screen.
Requires 16K RAM, has smooth scrolling, fast response with good sound effects.

METEOR SWARM (SV ONLY)

Another machine code game where your ship is caught in a meteor belt and you must blast your way out. Other ships are caught in the belt and will shoot at you if you do not destroy them first. Comes with bonus game - Birds of Orion.



SPECTRA-TRIVIA (SV 32K)
COMPU-TRIVIA (MSX)

The famous trivia game now on your computer. These are trivia quiz games for up to four players. The computer rolls the dice and each player moves in turn. At certain points you will be asked questions. There is provision for you to create your own question and answer file.

PLEASE SPECIFY SV318/328 OR MSX WITH ORDER.

>>>>>>>> **MSX** >>>>>>>>>>>>>>> **CARTRIDGES** <<<<<<<<<<<<<<< **MSX** <<<<<<<<

We have had requests from various people asking where they can buy CARTRIDGES so after a bit of searching around we came up with this list below which is for sale to subscribers.

Because of the price of the CARTRIDGES we do not hold them in stock, if you wish to buy them please send in your **MONEY AND ORDER** to us and we will get them in for you.

Prices include postage in Australia. Overseas add $2.00per Cartridge

TITLE	PRICE
MUSIC STUDIO	$68.00
GRAPHIC MASTER	"
PRINT LAB	"
MSX HOME WRITER (FOR PRINTER)	$41.00
TRAFFIC	$36.00
CRAZY TRAIN	$28.00
MOUSER	"
ALI BARBER & THE 40 THEIVES	"
DORODON	"
CHOPLIFTER	"
LOADRUNNER	"
RAID ON BUNGLING BAY	"
COSMO EXPLORER	"
PAYLOAD	"
ANTARCTIC ADVENTURE	"
MONKEY ACADEMY	"
SUPER SOCCER	$25.00
BACKGAMMON	"
ALPHA SQUADRON	"
SCION	"
CHARACTER COLLECTION I (PRINTER SOFTWARE)	$20.00
CHARACTER COLLECTION II " "	"

PLEASE ALLOW 14 DAYS FOR DELIVERY

COPY.COM patch for fast switch-off.

by Ross Moore.

If you have installed fast switch-off on a CP/M disk using Basic "Turnoff" program (Vol.2, No.1 P5), COPY.COM (used with 2 drives) won't work, as the program doesn't switch the drives on again!

The following patch partially corrects the problem, but only leaves the drives running for one copy run. (For more than 1, you have to exit COPY.COM and run it again.)

Place a CP/M disk with DDT.COM (or DDT80.COM for those fortunate enough to have an 80 Collumn Video Board, SV.806) and COPY.COM into drive A. Boot it up and type the following:-

(0100 is the Hex address, XX stands for the Hex value to be replaced, C3 is the Hex value replacing XX and <CR> is Enter or Carriage Return. **Caution**, read through this article before making any changes.)

```
A>DDT COPY.COM    <CR>
- S100            <CR>
0100 XX  C3       <CR>
0101 XX  50       <CR>
0102 XX  05       <CR>
0103 XX  .        <CR>

- S550            <CR>
0550 XX  3E       <CR>
0551 XX  07       <CR>
0552 XX  32       <CR>
0553 XX  50       <CR>    **  { See NOTE }
0554 XX  F0       <CR>    **  { below.    }
0555 XX  31       <CR>
0556 XX  63       <CR>
0557 XX  05       <CR>
0558 XX  C3       <CR>
0559 XX  03       <CR>
055A XX  01       <CR>
055B XX  .        <CR>

- ^C              <CR>
A> SAVE 5 COPY.COM
```

You now have a modified version of COPY.COM.(It is always advisable to keep a copy of the original program untouched. I do this by renaming the program with an extention of .OLD inplace of .COM ...MAX.)

NOTE **

This patch is for CP/m V2.22 (F-keys displayed). If you have CP/M V2.20, change 50 to 70 and F0 to EF.

COMPUTER JARGON

FLOPPY DISK

A thin flexible magnetic disk and a semi rigid protective jacket, in which the disk is permanently enclosed. There are 3 sizes of floppy disk 8", 5-1/4" and 3-1/2" and these disks may store data on one or both sides, at either single or double density. Depending on it's size the disk may store between 100 kilobytes to 1 megabyte.

FLOWCHART

A flowchart is a pictorial representation of an algorithm. The construction of a flowchart is one of the first steps in writing a program. Plastic templates with symbols which represent various operations are used to make up a flowchart, the name suggests that the operations flow into one another and it is a chart that is easily understood.

FOREGROUND PROGRAM

A program that has the highest priority and therefor takes precedence over other programs that are running at the same time in a multiprogramming environment.

FORTRAN

FORmula **TRAN**slator or Fortran as we call it is a high level programming language and was used principally by engineers and scientists, but nowdays many people know more than one language.

HALF DUPLEX

Transmission that takes place between a low speed terminal and a computer one way at a time on a two way circuit.

HANDSHAKING

The exchange of predetermined signals when a connection is first made across an interface, in order to confirm it is working properly and there is no data loss.

HARD DISK

A rigid storage disk which is capable of storing more information and can be accessed faster than a floppy disk.

INITIALIZATION

This is the process of setting up all variables to be used in a program at the start of the program.

INPUT

A signal that is transmitted from a peripheral device to the central processing unit.

INTEGRATED CIRCUIT (IC)

A combination of interconnected circuit elements placed on a motherboard, one IC may contain anywhere from a few to several thousands of transistors, resistors, diodes and capacitors.

INTERACTIVE

A conversational type system in which a continuous dialog can take place between a user and the computer,

LINE PRINTER

A peripheral device that prints an entire line of characters at one time, at speeds of 500-2000 lines per minute.

LIQUID CRYSTAL DISPLAY (LCD)

A visual display used in calculators and some lap-top computers, made of two glass plates sandwiched together with a special fluid.

LOADER

A program designed to load other programs into main memory usually from disk or tape.

MACHINE LANGUAGE

A language providing programs that can be expressed directly in a binary format acceptable to the central processing unit.
All other high level languages have to be translated into binary format before the CPU can read them.

MAGNETIC DISK

A flat disk with a magnetizable surface layer on which data can be stored by magnetic recording.
There are two main types of disk:

HARD DISK The hard disk rotates at high speed 2400rpm so that data may be accessed without due delay.
FLOPPY DISK - Which has been already explained.

MAINFRAME

A name that is given to any large central processing unit where terminals usually run off it.

NANOSECOND

A measurement of time which equals 1/1000 of a microsecond.

OBJECTCODE

The code of a users program after it has been translated.

OVERFLOW

In the result of an arithmetic operation if the answer is greater than the value that can be stored in that location.

CHIN

For MSX Computers

Can you solve this CHINESE MYSTERY?

This puzzle is solved when you can slide the LARGEST shape out at the foot of the board.

Use the Cursor Arrows to move the Selector hand over the pieces. Pressing the Space-bar along with a Cursor key moves a shape. The Maximum number of moves allowed is 300.

Place the HAND over BACK and press the Space-bar to retrace moves. Place the HAND over QUIT to finish the game.

```
10 SCREEN0:KEYOFF:WIDTH37:CLEAR300,
&HE6FF:DEFINTA-Z:COLOR15,4,1:SCREEN
0:ST=0:BC=10:MS=4:GOSUB670:COLOR1:G
OSUB540:GOSUB420
20 IFDT<0THEN240
30 J=STICK(ST):IFTGTHEN90
40 IFJ=3THENIFCX+CY=295THEN240
50 X=CX+DX(J)*MS:Y=CY+DY(J)*MS
60 IFX>31ANDX<145THENCX=X
70 IFY>6ANDY<152THENCY=Y
80 GOTO20
90 X=CX\32:Y=(CY+25)\32
100 P=Y*6+X:N=B(P):IFN<>B(((CY+40)\
32)*6+(CX+15)\32)THEN50
110 IF(JMOD2)=0THEN20
120 IFN=10THEN20
130 IFN=B(P-6)THENY=Y-1:P=P-6
140 IFN=B(P-1)THENX=X-1:P=P-1
150 R=J\2:IFB(P+F0(N,R))<>10THEN20
160 IFB(P+F1(N,R))<>10THEN20
170 A=(Y+(R=0))*1024+(X+(R=3))*32-7
68:BX=SX(N)-(R=1)-(R=3):BY=SY(N)-(R
=0)-(R=2):FORI=1TO4:DM=USR0(A)+USR1
(BX)+USR2(BY)+USR3(R):CX=CX+DX(J)*8
:CY=CY+DY(J)*8:NEXT:Q=(Y+DY(J))*6+(
X+DX(J)):FORY=0TOSY(N)
180 FORX=0TOSX(N):B(P+Y*6+X)=10:NEX
T:NEXT:FORY=0TOSY(N):FORX=0TOSX(N):
B(Q+Y*6+X)=N:NEXT:NEXT:BM(T)=Q*10+(
(J+4)MOD8):T=T+DT:INTERVALOFF
190 LINE(182,20)-(217,35),7,BF:DRAW
"BM185,25":PRINT#1,RIGHT$("   "+STR
$(T),4):INTERVALON
200 IFQ=26ANDN=9THEN370
210 IFT<MXTHEN60
220 INTERVALOFF
230 LINE(182,20)-(217,35),8,BF:DRAW
"BM185,25":PRINT#1,"300!":PLAY"o1da
d":INTERVALON
240 CX=210:CY=143
250 J=STICK(ST):IFTGANDCY=79THEN380
260 IFTGANDCY=143THEN350
270 IFJ=0THEN250
280 IFJ=7ANDT<MXTHEN320
290 IFJ=1ANDCY=143THENCY=79
300 IFJ=5ANDCY=79THENCY=143
310 GOTO250
320 CX=144:CY=151:DT=1
330 IFSTICK(ST)THEN330
340 GOTO20
350 IFT=0THEN250
360 PLAY"VT200OF16A16":J=BM(T-1)MOD
10:P=BM(T-1)\10:X=PMOD6:Y=P\6:CX=X*
32+8:CY=Y*32-17:DT=-1:GOTO100
370 LINE(64,176)-(127,191),BC,BF:FO
RI=1TO3:POKE&HE7C9,11:D=USR0(3392)+
USR1(1)+USR3(2):NEXT:DRAW"BM70,112"
:PRINT#1,"WIN":CX=110:CY=106:GOSUB4
00
380 INTERVALOFF
390 PLAY"VT200L8O4E4O3BO4DEABO5DE2"
:COLOR15:CLS:END
400 TG=STRIG(ST):IFTGTHENC=8ELSEC=1
4
410 PUTSPRITE0,(CX,CY),1:PUTSPRITE1
,(CX,CY),C:RETURN
420 DIMB(41),F0(9,3),F1(9,3):DIMSX(
9),SY(9),DX(8),DY(8):MX=300:DIMBM(M
X):FORI=0TO41:B(I)=-1:NEXT:B(32)=10
:B(33)=10
430 RESTORE440:FORN=0TO9:READP,SX(N
),SY(N):FORY=0TOSY(N):FORX=0TOSX(N)
:B(P+Y*6+X)=N:NEXT:NEXT:FORR=0TO3:R
EADF0(N,R),F1(N,R):NEXT:NEXT
440 DATA7,0,1,-6,-6,1,7,12,12,-1,5,
19,0,1,-6,-6,1,7,12,12,-1,5,10,0,1,
-6,-6,1,7,12,12,-1,5,22,0,1,-6,-6,1
,7,12,12,-1,5,20,1,0,-6,-5,2,2,6,7,
-1,-1,31,0,0,-6,-6,1,1,6,6,-1,-1,26
,0,0,-6,-6,1,1,6,6,-1,-1,27,0,0,-6,
-6,1,1,6,6,-1,-1
450 DATA34,0,0,-6,-6,1,1,6,6,-1,-1,
8,1,1,-6,-5,2,8,12,13,-1,5
460 PLAY"VT200L8O4E4O3BO4DEABAE2":R
ESTORE480:FORJ=0TO1:S$="":FORI=0TO3
1:READD$:S$=S$+CHR$(VAL("&H"+D$))
470 NEXT:SPRITE$(J)=S$:NEXT
480 DATA0,0,2,5,5,5,35,2D,24,20,10,
C,4,4,0,0,0,0,A0,50,50,58,54,54,4,4
,8,10,10,10,0,0,0,0,0,0,2,2,2,2,12,1B
,1F,F,3,3,3,0,0,0,0,0,A0,A0,A0,A8,A
8,F8,F8,F0,E0,E0,E0,0,0
490 ONINTERVAL=3GOSUB400
500 CX=112:CY=55:INTERVALON
510 RESTORE520:FORI=1TO8:READDX(I):
NEXT:FORI=1TO8:READDY(I):NEXT
520 DATA0,1,1,1,0,-1,-1,-1,-1,-1,0,
1,1,1,0,-1
530 T=0:DT=1:RETURN
```

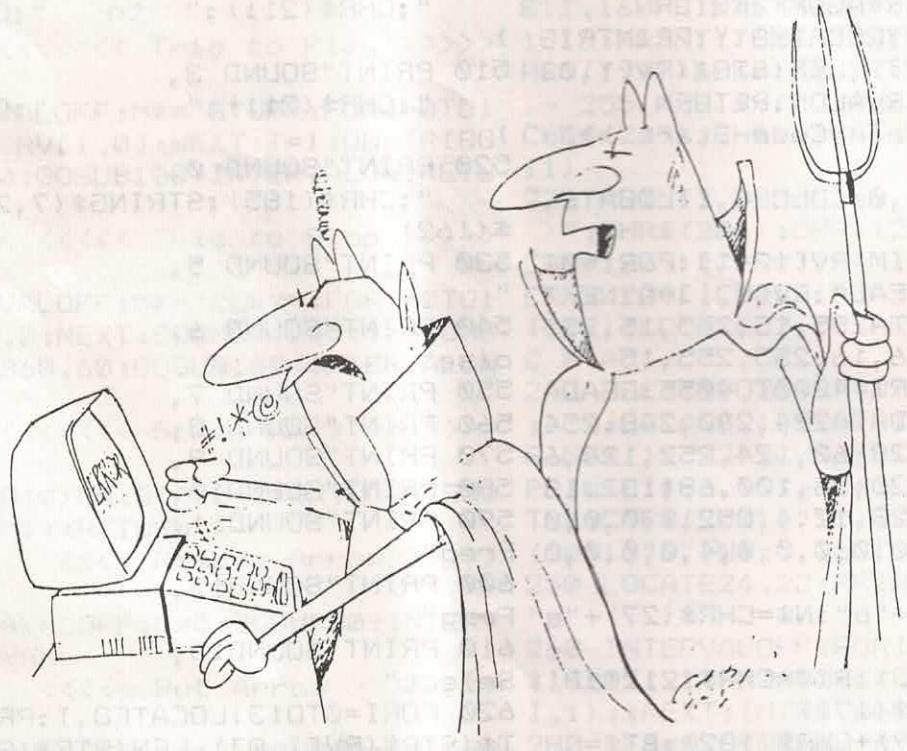
```
540 SCREEN2,2,0:OPEN"GRP:"FOROUTPUT
AS#1:LINE(0,0)-(255,191),1,BF:LINE(
24,0)-(167,175),12,BF:LINE(32,8)-(1
59,167),BC,BF:LINE(64,168)-(127,175
),BC,BF:DRAW"A0S4C1BM128,175U7R32U1
61L129D161R32D7":RESTORE570
550 FORY=20TO148STEP64:IFY=20THENC=
7ELSEC=15
560 LINE(182,Y)-STEP(35,15),C,BF:DR
AW"C1BM-32,-10":READD$:PRINT#1,D$:N
EXT
570 DATA"0000","QUIT","BACK"
580 DRAW"BM35,8R25F3D57G3L25H3U57E3
BR6BD42F2RFR6ERE4UE3UG2UE2U7H4U2HLH
LHL7GLGLGD2GDG2DGD5FD3F3RU3H2U3R2U3
E5R2F2E2R2F4D4RFBU4BL5HL2GBL3HL2GBD
3ER3DGLURBR4ER3DGLURBD3BL3D2LBR3BD2
GL5BU3BLGBU18F5U5BM35,72R25F3D57G3L
25H3U57"
590 DRAW"E3BR3BD37U3L3D4F2RFDF3RFR8
ERE3UEUE2U3H3D5U2E3U5HLHUH2UH2LHL8G
LGDGD3L2DGDGD6R4U2EU2EUE2RER5DRUR3F
2D3FDBL2H2L3DR3BL9R3UL3G2BD3E2R2FDG
LUHR2DBR4E2RFDGLU2RDBD5GDL2GL3GEU2B
R4BU3DLBL6HLGDFREBU16BE2F2RUBM131,8
R25F3D57G3L25H3U57E3BR10BD19GLG3D2H
D6U4LD9U3"
600 DRAW"L2GD3F2D2FDF3RFR8ERE3UEU2E
2U3HLU4HUEUHUH5L5D3RER2F4DGDFD2FD2F
D2FD2BU6BL4L4ER2BL9R2FL4BD3FREL4ER3
FBR7RHL3GDR2ULBL4D2G2D3R3BD2BL2R4GL
2DR2E3HUF3BL10UBM131,72R25F3D57G3L2
5H3U57E3BR4BD35U4L2GD4F2RDGD3FDF2RF
R8ERER"
610 DRAW"EUEU3HURE2U4HL2D4U9HUHUH3L
HUH2LGLR3FL3FR2FL6DLG2D8LU6E2D6ER3U
L3U6BR9FD5GL2UR4U4FD3RDL2FRD3BL2D3G
L2HR3UL3U2ER2FBL7D3GL2HR3UL3U2ER2BR
2BD7L2GD2FR3EBR2BD3D2HL6U2BM67,8R57
F3D57G3L57H3U57E3BR18BD39ED2FDF3RFR
5ERERE2UEU2FER2EU9HUHUHLHU2H2LH2L7"
620 DRAW"GLGL2GHLGDFGEREFR2U2D5L3U2
D2L3UDGLD2GD9FR3HU4R3D2U4EUE2RERFRF
RE2RFRF2DFD2R3D4BL3BU2ULHL4GRDR3ULB
U3BR2HL3GBL4HL3GBLBD3ERD2RU2R2DGREF
DGDFBL2BD2FR3ERBU16L3E5"
630 PAINT(92,25):PSET(89,33),BC:DRA
W"C1BM67,72R57F3D25G3L57H3U25E3BR15
BD18UH2LGD2F3RDF2R2FR16ER2E2URE3U2H
L2GDGEUE2UGU3H2LHL2HL5U3LD4LU4LD4LU
2GL4GL2D3GU3LG2D4HUR2U2ED4FRFRDFR2U
2RD2REU2HL4G"
640 DRAW"BU4R4F2BD3R4DFRU2RD2REU2HL
3GEHER2ER2BD4RERE2U2HD3HU2HD2BD11BL
6LHL5GLBM35,136":GOSUB650:DRAW"BM67
,104":GOSUB650:DRAW"BM99,104":GOSUB
650:DRAW"BM131,136"
650 DRAW"R25F3D25G3L25H3U25E3BRBD17
GDF2R2DF3R2FR6ER2E3URE2U2H3D3E2U3HD
3HU3H2LHL2DGU4HD5LU5LD5HU3DGLGLG2LG
2D5FU6ED6U4RU2BR2BDER3FBR3ER3FBD3BL
L4D2R3UL2BL5L2DR3U2L5BD5BRFRFR4EREB
UBL4L"
660 RETURN
670 LOCATE2,1:PRINT"CAN YOU SOLVE T
HIS CHINESE MYSTERY  ";STRING$(34,
"━")SPC(77)"THIS PUZZLE IS SOLVED W
HEN YOU CAN"SPC(40)"SLIDE THE LARGE
ST SHAPE OUT AT THE"SPC(40)"THE FOO
T OF THE BOARD.":PRINT:PRINT
680 PRINT"  USE THE CURSOR ARROWS T
O MOVE THE":PRINT:PRINT"  SELECTOR
HAND OVER THE PIECES."SPC(81)"PRESS
ING THE SPACE BAR ALONG WITH"SPC(41
)"A CURSOR KEY MOVES A SHAPE."SPC(8
8);:PRINT"PRESS ANY KEY TO CONTINUE
":BEEP
690 A$=INKEY$:IFA$=""THEN690ELSEBEE
P
700 CLS:LOCATE2,3:PRINT"THE MAXIMUM
 NUMBER OF MOVES ALLOWED"SPC(39)"IS
 300."SPC(141)"PLACE THE HAND OVER
BACK AND PRESS"SPC(23)"━━━━"SPC(13)
"THE SPACE BAR TO RETRACE MOVES."SP
C(117);
710 PRINT"PLACE THE HAND OVER QUIT
TO FINISH"SPC(23)"━━━━"SPC(13)"THE
GAME."SPC(145);:PRINT"PRESS ANY KEY
 TO START"
720 A$=INKEY$:IFA$=""THEN720ELSECLS
730 RESTORE750:FORA=&HE700TO&HE7C3:
READD$:D=VAL("&H"+D$)
740 POKEA,D:NEXT:DEFUSR0=&HE700:DEF
USR1=&HE707:DEFUSR2=&HE714:DEFUSR3=
&HE71E:RETURN
750 DATA2A,F8,F7,22,C4,E7,C9,2A,F8,
F7,23,29,29,29,29,29,22,C7,E7,C9,3A
,F8,F7,3C,87,87,32,C9,E7,C9,3A,F8,F
7,32,C6,E7,CD,2E,E7,CD,6A,E7,CD,4E,
E7,C9,2A,C4,E7,11,2A,E8,ED,4B,C7,E7
,3A,C9,E7,F5,C5,D5,E5,CD
760 DATA59,0,E1,24,D1,C1,EB,9,EB,F1
,3D,20,EE,C9,ED,5B,C4,E7,ED,4B,C7,E
7,3A,C9,E7,F5,C5,D5,E5,CD,5C,0,E1,D
1,C1,9,14,F1,3D,20,F0,C9,3A,C6,E7,B
7,28,4E,3D,28,28,3D,28,18,21,32,E8,
E5,78,2F,47,79,2F,4F
770 DATA3,62,6B,9,E5,21,F8,FF,9,44,
4D,E1,18,1B,21,2A,E8,ED,42,EB,ED,42
,D5,ED,B0,E1,C9,21,F8,FF,9,E5,11,2A
,E8,19,C1,11,22,E8,D5,3A,C9,E7,8,C5
,1,8,0,ED,B0,C1,9,EB,9,EB,8,3D,20,F
0,E1,C9,21,2A,E8,ED,B0,C9
```

```
            ************
            * SOUND.ED *
            ************
         for the SV 318/328
 A program to let you explore the
capabilities of the Programmable
Sound Generator.
 Use the joystick (or arrow keys)
to set up and alter the register
values. You may change the
contents of Registers 7 & 13 with
numeric keys.
 Note that after changing the
contents of Register 13, you must
stop & restart the sound (using
the trigger or space bar) to hear
the effect.
 Press ESC (from Reg 7 or 13) or
^STOP to end.
 Press 'R' (from Reg 7 or 13) for
 Restart.

10 STOPON:ONSTOPGOSUB830:GOTO390
20 '<<<<<<<<<<< Flash Arrow >>>>>>
>>>>
30 L=1-L:LOCATEX,Y:IFLTHENPRINT" "
;ELSEPRINTLE$
40 RETURN
50 '<<<<<<<<<<< Trig to Play >>>>>
>>>>
60 INTERVALOFF:M$="STOP":FORI=0TO1
3:SOUNDI,RV[I,0]:NEXT:T=1:ONSTRIGG
OSUB80,80:GOSUB100:INTERVALON:RETU
RN
70 '<<<<<<<<<<< Trig to Stop >>>>>
>>>>
80 INTERVALOFF:M$="PLAY":FORI=0TO1
3:SOUNDI,0:NEXT:SOUND7,255:T=0:ONS
TRIGGOSUB60,60:GOSUB100:INTERVALON
:RETURN
90 '<<<<<<<<<<< Strig On >>>>>>>>>
>>>>
100 STRIG(0)ON:STRIG(1)ON:LOCATE30
,3:PRINTM$;:RETURN
110 '<<<<<<<<<< Remove Arrow >>>>>
>>>>
120 INTERVALOFF:L=0:GOSUB30:INTERV
ALON:RETURN
130 '<<<<<<<<<< Put Arrow >>>>>>>>
>>>>
140 INTERVALOFF:L=1:GOSUB30:INTERV
ALON:RETURN
150 '<<<<<<<< Print Lower Screen >
>>>>
160 INTERVALOFF:LOCATE0,14:PRINTCH
R$(27)+"J":ONY+1GOTO170,190,170,19
0,170,190,210,220,270,270,270,290,
290,310
170 IFY=0THENCH$="A"ELSEIFY=2THENC
H$="B"ELSEIFY=4THENCH$="C"
180 LOCATE3,15:PRINT"FREQUENCY CHA
NNEL ";CH$;", FINE ADJUST":LOCATE3
,16:PRINTSTRING$(32,174):LOCATE4,1
8:PRINT"Decrease  ";LE$;" 0 - 255
";RI$;"  Increase":INTERVALON:RETU
RN
190 IFY=0THENCH$="A"ELSEIFY=2THENC
H$="B"ELSEIFY=4THENCH$="C"
200 LOCATE3,15:PRINT"FREQUENCY CHA
NNEL ";CH$;", COARSE ADJUST":LOCAT
E3,16:PRINTSTRING$(34,174):LOCATE4
,18:PRINT"Decrease  ";LE$;" 0 - 25
5 ";RI$;"  Increase":INTERVALON:RE
TURN
210 LOCATE10,15:PRINT"NOISE FREQUE
NCY":LOCATE10,16:PRINTSTRING$(15,1
74):LOCATE2,18:PRINT"Decrease  ";L
E$;" 0 - 31 ";RI$;"  Increase":INT
ERVALON:RETURN
220 LOCATE11,15:PRINT"MIXER":LOCAT
E11,16:PRINTSTRING$(5,174):LOCATE2
1,16:PRINT"TONE ";STRING$(8,209);C
HR$(194):LOCATE4,17:PRINTLE$;" 192
 - 255 ";RI$;"   NOISE "CHR$(209);
CHR$(209);CHR$(194);"     ";CHR$(2
11)
230 PRINT" Decrease   Increase
  ";CHR$(206);CHR$(209);CHR$(183);
CHR$(209);CHR$(194);" ";CHR$(206);
CHR$(209);CHR$(183);CHR$(209);CHR$
(194):LOCATE22,19:PRINT"X X C B A
C B A"
240 LOCATE9,20:PRINT"or           "
;BT$:LOCATE21,21:PRINTCHR$(211);ST
RING$(15,32);CHR$(211):LOCATE1,21:
PRINTB1$;"   ";B1$:LOCATE1,22:PRIN
TR$;CHR$(200);"1";N$;" - ";R$;CHR$
(200);"6";N$;" To Toggle   ";BB$;
250 LOCATE24,23:PRINT"0=yes - 1=No
";
260 INTERVALOFF:FORI=1TO8:LOCATE(2
*I+20),21:PRINTMID$(BIN$(RV[7,0]),
I,1);:NEXT:INTERVALON:RETURN
270 IFY=8THENCH$="A"ELSEIFY=9THENC
H$="B"ELSEIFY=10THENCH$="C"
```

```
280 LOCATE9,15:PRINT"VOLUME CHANNE
L ";CH$;:LOCATE9,16:PRINTSTRING$(1
6,174):LOCATE2,18:PRINT"Decrease
";LE$;" 0 - 16 ";RI$;"  Increase":
LOCATE6,20:PRINT"0-15: Tone Volume
":LOCATE8,21:PRINT"16: Volume Dete
rmined by SHAPE":INTERVALON:RETURN
290 IFY=11THENCH$="FINE ADJUST  ":
LE=29ELSECH$="COARSE ADJUST":LE=31
300 LOCATE4,15:PRINT"ENVELOPE PERI
OD - ";CH$;:LOCATE4,16:PRINTSTRING
$(LE,174):LOCATE4,18:PRINT"Decreas
e  ";LE$;" 0 - 255 ";RI$;"  Increa
se":INTERVALON:RETURN
310 LOCATE10,15:PRINT"ENVELOPE SHA
PE":LOCATE10,16:PRINTSTRING$(4,174
);CHR$(217);CHR$(216);STRING$(8,17
4)
320 LOCATE2,17:PRINTS1$:LOCATE2,16
:PRINTB1$:LOCATE26,16:PRINTB1$
330 LOCATE2,19:PRINTS2$:LOCATE2,21
340 LOCATE2,18:PRINTB1$:LOCATE14,1
8:PRINTB1$:LOCATE26,18:PRINTB1$
350 LOCATE2,21:PRINTS3$:LOCATE5,23
:PRINT"Press 0 - 7 to Select";
360 LOCATE2,20:PRINTB1$:LOCATE14,2
0:PRINTB1$:INTERVALOFF:RETURN
370 INTERVALOFF:LOCATE8,Y:PRINTRIG
HT$(STR$(RV[Y,0]),LEN(STR$(RV[Y,0]
))-1);"  ":INTERVALON:RETURN
380 '<<<<<<<<< Main Code Start >>>>
>>>>
390 CLS:SCREEN0,0:COLOR4,1:LOCATE,
,0
400 DEFINTA-Z:DIM RV[13,1]:FORI=0T
013:RV[I,0]=0:READA:RV[I,1]=A:NEXT
:RV[7,0]=255:DATA255,15,255,15,255
,15,31,255,16,16,16,255,255,15
410 X=12:Y=0:FORI=4000TO4055:READA
:VPOKEI,A:NEXT:DATA224,240,248,254
,248,240,224,0,28,60,124,252,124,6
0,28,0,4,12,28,20,36,100,68,132,13
2,68,100,36,20,28,12,4,252,0,0,0,0
,0,0,252,252,0,0,0,0,0,0,4,0,0,0,0
,0,0,0,4
420 R$=CHR$(27)+"p":N$=CHR$(27)+"q
"
430 LE$=CHR$(213):RI$=CHR$(212):B1
$=CHR$(218)+CHR$(171)
440 BT$=CHR$(209)+CHR$(182):BT$=CH
R$(176)+BT$+BT$+BT$+BT$+BT$+BT$+BT
$+CHR$(209)+CHR$(164):BB$=CHR$(209
)+CHR$(183):BB$=CHR$(185)+BB$+BB$+
BB$+BB$+BB$+BB$+BB$+CHR$(209)+CHR$
(162)
450 FORI=1TO45:READA:S1$=S1$+CHR$(
A):NEXT:DATA27,112,200,48,27,113,3
2,190,171,171,171,171,171,32,32,32
,27,112,200,49,27,113,32,214,171,1
71,171,171,171,32,32,32,27,112,200
,50,27,113,200,215,215,215,215,215
,215
460 FORI=1TO45:READA:S2$=S2$+CHR$(
A):NEXT:DATA27,112,200,51,27,113,2
00,190,188,190,188,190,188,32,32,3
2,27,112,200,52,27,113,32,215,174,
174,174,174,174,32,32,32,27,112,20
0,53,27,113,32,214,214,214,214,214
,214
470 FORI=1TO29:READA:S3$=S3$+CHR$(
A):NEXT:DATA27,112,200,54,27,113,3
2,188,174,174,174,174,174,32,32,32
,27,112,200,55,27,113,32,188,190,1
88,190,188,190
480 PRINT"SOUND 0,          Freq A
  ";CHR$(176);STRING$(7,209);CHR
$(164)
490 PRINT"SOUND 1,          Freq A
  ";CHR$(211);"Trigger";CHR$(211
)
500 PRINT"SOUND 2,          Freq B
  ";CHR$(211);"  to   ";CHR$(211
)
510 PRINT"SOUND 3,          Freq B
  ";CHR$(211);"       ";CHR$(211
)
520 PRINT"SOUND 0,          Freq C
  ";CHR$(185);STRING$(7,209);CHR
$(162)
530 PRINT"SOUND 5,          Freq C
"
540 PRINT"SOUND 6,          Freq N
oise"
550 PRINT"SOUND 7,          Mixer"
560 PRINT"SOUND 8,          Vol A"
570 PRINT"SOUND 9,          Vol B"
580 PRINT"SOUND10,          Vol C"
590 PRINT"SOUND11,          Shape
Freq"
600 PRINT"SOUND12,          Shape
Freq"
610 PRINT"SOUND13,          Shape
Select"
620 FORI=0TO13:LOCATE8,I:PRINTRIGH
T$(STR$(RV[I,0]),LEN(STR$(RV[I,0])
)-1):NEXT
630 GOSUB160
640 ONINTERVAL=25GOSUB30:INTERVALO
N
```

```
650 GOSUB80
660 ONSTRIGGOSUB60,60:STRIG(0)ON:S
TRIG(1)ON
670 S=STICK(0)+STICK(1)
680 IFS<>1ANDS<>5THEN720
690 IFS=1ANDY>0THENGOSUB120:Y=Y-1:
GOSUB160:GOSUB130ELSEIFS=1THENGOSU
B120:Y=13:GOSUB160:GOSUB130
700 IFS=5ANDY<13THENGOSUB120:Y=Y+1
:GOSUB160:GOSUB130ELSEIFS=5THENGOS
UB120:Y=0:GOSUB130:GOSUB160
710 FORI=1TO50:NEXT:GOTO670
720 IFY<>7ANDY<>13THEN770
730 A$=INKEY$:IFA$=" "THENGOTO730E
LSEIFA$=CHR$(27)THEN830ELSEIFA$="R
"THENRUN
740 IFY=7ANDA$>"0"ANDA$<"7"THENIFM
ID$(BIN$(RV[7,0]),(-VAL(A$)+9),1)=
"0"THENRV[7,0]=RV[7,0]+(2^(VAL(A$)
-1)):GOSUB260:GOSUB370ELSERV[7,0]=
RV[7,0]-(2^(VAL(A$)-1)):GOSUB260:G
OSUB370:GOTO810
750 IFY=13ANDA$>="0"ANDA$<"8"THENR
ESTORE760:FORI=0TOVAL(A$):READA:NE
XT:RV[13,0]=A:GOSUB370
760 DATA0,4,8,10,11,12,13,14
770 IFS=3THENIFRV[Y,0]<RV[Y,1]THEN
RV[Y,0]=RV[Y,0]+1:GOSUB370ELSERV[Y
,0]=0+192*-(Y=7):GOSUB370
780 IFS=3ANDY=7THENGOSUB260
790 IFS=7THENIFRV[Y,0]>0+192*-(Y=7
)THENRV[Y,0]=RV[Y,0]-1:GOSUB370ELS
ERV[Y,0]=RV[Y,1]:GOSUB370
800 IFS=7ANDY=7THENGOSUB260
810 IF T=1ANDY<>13THENSOUNDY,RV[Y,
0]
820 FORI=1TO100:NEXT:GOTO670
830 INTERVALOFF:COLOR2,1:SCREEN0,1
:LOCATE,,1:CLS:END
```



"Calm down Edgar—the guy who thought up the damn thing must be down here somewhere."

CPMCSR.BAS

```
100 :'              CPMCSR.BAS
110 :'                by TIL
120 :'This program checks keyboard
map flag
130 :'on system track of CP/M 2.20
disk and
140 :'changes flag to allow curaor
keys to
150 :'be used as ^E, ^S, ^D AND ^X.
160 :'i.e. WORDSTAR, dBASE etc. key
s.
170 DEFINT A-Z
180 CLEAR 1000
190 :'SET UP RANDOM BUFFER
200 FIELD #0,128 AS A$,128 AS B$
210 OUT &H34,0
220 CLS:PRINT"CP/M CURSOR REDEFINIT
ION.":PRINT
230 PRINT"Use only blank & SYSGENed
"
240 PRINT"or COPYed CP/M 2.20 disks
.":PRINT
250 PRINT"Insert CP/M disk in Drive
 #1"
260 PRINT"and press ENTER to contin
ue...."
270 IF INKEY$<>CHR$(13) THEN 270
280 :'Read first BIOS system sector
 at
290 :'Track 0 Sector 2 which contai
ns
300 :'the keyboard map flag.
310 C$=DSKI$(1,0,2)
320 :'Read cold boot jump address
330 VER$=MID$(A$,2,2)
340 :'CP/M 2.20 cold boot jump
350 :'address is &HEE10
360 ACPM$=CHR$(&H10)+CHR$(&HEE)
370 IF VER$<>ACPM$ THEN 380 ELSE 41
0
380     PRINT"Use only bootable CP/M
 2.20 disk."
390     PRINT"Press ENTER to try aga
in.":PRINT
400     IF INKEY$=CHR$(13) THEN 210
ELSE 400
410 PRINT"SV CP/M 2.20 disk."
420 :'Read kybd map flag
430 FLAG$=MID$(A$,97,1)
440 IF FLAG$=CHR$(0) THEN 450 ELSE
470
450     PRINT"Cursor keys are alread
y"
460     GOTO 600
470 PRINT"SV BASIC type cursor keys
.":PRINT
480 PRINT"Convert to CTRL ESDX type
 keys?"
490 PRINT"Confirm  (Y/N)?"
500 CNFRM$=INKEY$:IF CNFRM$<>"" THE
N 510 ELSE 500
510 IF CNFRM$="Y" OR CNFRM$="y" THE
N 560 ELSE 520
520     IF CNFRM$="N" OR CNFRM$="n"
THEN 530 ELSE 500
530         PRINT"---ABORT---"
540         GOTO 610
550 :'Set kybd map flag.
560 MID$(A$,97,1)=CHR$(0)
570 :'Write back modified sector.
580 DSKO$ 1,0,2
590 PRINT"Cursor keys converted to"
600 PRINT"CTRL ESDX type cursor key
s."
610 OUT &H34,0
620 PRINT:PRINT"Remove CP/M disk fr
om drive."
630 END
```

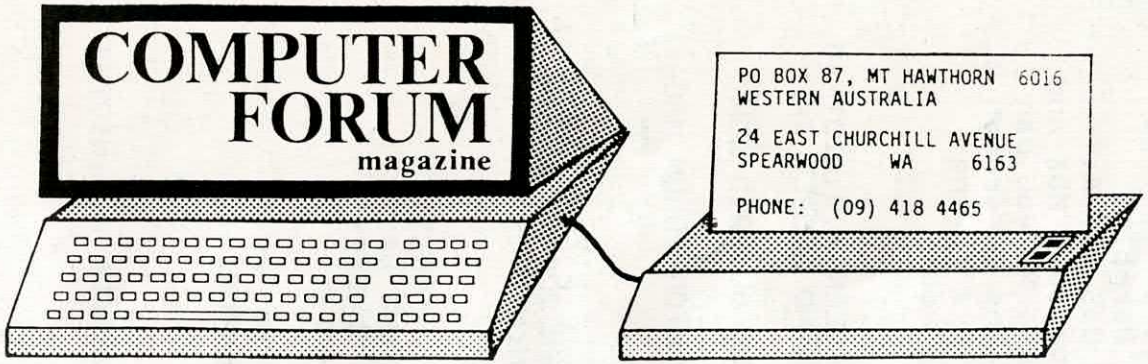
BANNER

The following program is from JOHN RUDZINSKI of Detroit, Michigan, in the U.S. of A.

It is suitable for both MSX and SV-318/328 computers. As you can see, this program runs on screen 0.

The program prints a moving Banner across the screen.

```
10 CLEAR1000
20 CLS:COLOR15,1:SCREEN 0,0:WIDTH40
30 B$="############ NO DOUBLE TALK
###########"
40 C$="********** WE DO GOOD WORK!
**********"
50 A$="======= COMPUTOR FORUM MAGAZ
INE ======="
60 LOCATE0, 8,0
70 PRINT A$
80 IF INKEY$<>""THEN 220
90 D$=RIGHT$(A$,LEN(A$)-1)
100 E$=LEFT$(A$,1)
110 A$=D$+E$
120 FOR I = 1 TO 99 : NEXT I
130 LOCATE0,14,0:PRINT B$
140 Y1$=RIGHT$(B$,LEN(B$)-1)
150 Z1$=LEFT$(B$,1)
160 B$=Y1$+Z1$
170 LOCATE0,20,0:PRINT C$
180 Y$=RIGHT$(C$,LEN(C$)-1)
190 Z$=LEFT$(C$,1)
200 C$=Y$+Z$
210 GOTO60
220 CLS:COLOR15,4:SCREEN 0,0:WIDTH4
0
230 END
```



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